region to constitute collector current I_C. In this way, almost the entire emitter current flows to the collector circuit. It is clear that emitter current is the sum of collector and base current, i.e. $I_E = I_B + I_C$.

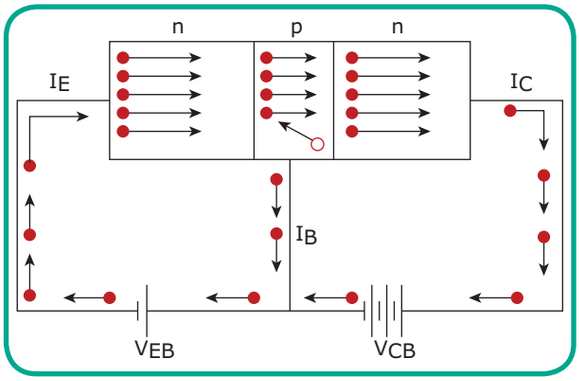


Figure 5.8 Working of NPN transistor

5.7.2 Working of PNP Transistor

Figure 5.9 shows the basic connection of a PNP transistor. The forward bias causes the holes in the p type emitter to flow towards the base. This constitutes the emitter current I_E. As these holes cross into n-type base, they tend to combine with electrons. As the base is lightly doped and very thin, only a few holes (less than 5%) combine with the electrons. The remainder holes (more than 95%) cross into the collector region to constitute the collector current I_C. In this way, almost the entire emitter current flows into the collector region. It may be noted that current conduction within PNP transistor is by holes. However, in the external connecting wires, the current is still denoted by electrons.

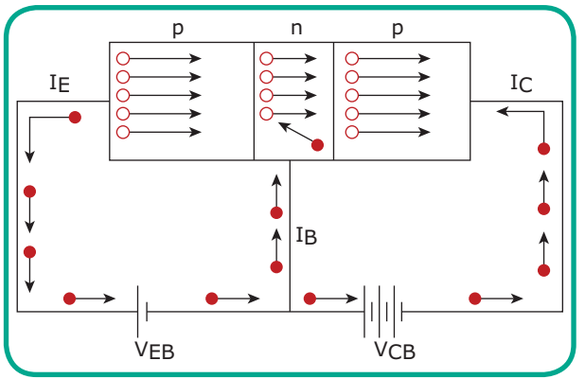


Figure 5.9 Working of PNP transistor

5.8 OPERATING MODES OF TRANSISTORS

Depending on the biasing conditions like forward or reverse, transistors have three major modes of operation namely, cut-off, active and saturation regions.

Active Mode: In this mode, transistor is generally used as a current amplifier. In active mode, two junctions are differently biased, that means, emitter-base junction is forward biased and collector-base junction is reverse biased. In this mode, current flows between emitter and collector and amount of current flow is proportional to base current.

Cut-Off Mode: In this mode, both collector-base junction and emitter-base junction are reverse biased. This in turn not allows the current to flow from collector to emitter, when the base-emitter voltage is low. In this mode, device is completely switched-off; as a result, the current flowing through the device is zero.

Saturation Mode: In this mode of operation, both the emitter-base and collector-base junctions are forward biased. Current flows freely from collector to emitter, when the base-emitter voltage is high. In this mode, device is fully in the switched ON condition.



Transistor acts as a switch.

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